



جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

IAU

COLLEGE OF ARCHITECTURE AND PLANNING

DEPT. OF BUILDING ENGINEERING

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Design Studio (6)

Technical Project

Structural Design
Load Calculation and Structural Element Design

Project Type
STUDENT DORM

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TWO-WAY RIBBED SLAB (7.2mx6.0m) (R1)

GIVEN:-

$$f_y = 420 \text{ MPa} \quad f_c = 35 \text{ MPa} \quad \Phi = 16 \text{ mm}$$

$$\Phi_{\text{stirrups}} = 10 \text{ mm} \quad WLL = 1.9 \text{ KN/m}^2$$

To find h

$$h = \frac{\ln(0.8 + \frac{f_y}{1500})}{36 + 5\beta(\alpha m - 0.2)}$$

Assumed $\alpha m = 2$

$$\beta = 7.2/6.0 = 1.2$$

$$h = 7.2(0.8 + 420/1500)/36 + 5 \times 1.2(2 - 0.2) = 166 \text{ mm}$$

$$h_{\text{used}} = 300 \text{ mm}$$

effective depth

$$d = h - c - \Phi/2 - \Phi_{\text{stirrups}} = 300 - 10 - 8 - 10 = 272 \text{ mm}$$

LOAD CALCULATIONS:-

Wdl=

$$\text{Upper slab(solid)} = (1 \times 1 \times 0.1) \times 24 = 2.4 \text{ KN/m}^2$$

$$\text{Ribs (x-direction)} = 2 \times (0.1 \times 0.2 \times 1) \times 24 = 0.96 \text{ KN/m}^2$$

$$\text{Ribs (y-direction)} = 2 \times (0.1 \times 0.2 \times 1) \times 24 = 0.96 \text{ KN/m}^2$$

$$\text{Blocks} = (8 \times 2) \times 0.15 = 2.4 \text{ KN/m}^2$$

$$\text{Slab self-weight per } 1 \text{ m}^2 = 6.72 \text{ KN/m}^2$$

Finishing=

$$\text{Porcelain} = (1 \times 1 \times 0.005) \times 21 = 0.105 \text{ KN/m}^2$$

$$\text{Mortar} = (1 \times 1 \times 0.05) \times 20 = 1 \text{ KN/m}^2$$

$$\text{Gypsum board ceiling} = (1 \times 1 \times 0.02) \times 4.4 = 0.08 \text{ KN/m}^2$$

$$\text{Total finishing load per } 1\text{m}^2 = 1.185 \text{ KN/m}^2$$

$$\text{TOTAL DEAD LOAD} = 6.72 + 1.185 = 7.905 \text{ KN/m}^2$$

ULTIMATE LOAD:

$$1.2(\text{WDL}) + 1.6(\text{WLL}) = 1.2(7.905) + 1.6(1.9) = 12.526 \text{ KN/m}^2$$

-UL PER RIB:

$$\text{UL/RIB} = 12.526/2 = 6.263 \text{ KN/m}^2$$

CALCULATION OF MOMENT:-

$$-M_u = CUL^2 = 0.033 \times 6.263(7.2^2) = 10.71 \text{ KN/RIB}$$

$$-M_n = M_u/0.9 = 10.71/0.9 = 11.9 \text{ KN/RIB}$$

STEEL RATIO:-

$$-R_n = M_n/1000 \times b(d^2) = 11.9/1000 \times 1 \times (0.272^2) = 0.16 \text{ MPa}$$

$$-m = f_y/0.85f_c = 420/0.85 \times 35 = 14.118$$

$$\rho = \frac{1}{m} \left[1 - \sqrt{1 - \frac{2mR_n}{f_y}} \right]$$

$$\rho = 1/14.118(1 - \sqrt{1 - 2 \times 14.118 \times 0.16/420}) = 3.82 \times 10^{-4}$$

$$\rho_{\min} = 1.4/f_y = 1.4/420 = 3.33 \times 10^{-3}$$

STEEL AREA

$$A_s = \rho b d = 3.33 \times 10^{-3} \times 500 \times 272 = 453.33$$

$$A_b = \pi d^2 / 4$$

$$= \pi (16)^2 / 4 = 201 \text{ mm}^2$$

Or

$$= \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

#n.bars

$$A_s / A_b$$

$$= 453.33 / 201 = 2.25$$

Or

$$453.33 / 254.5 = 1.84 = 2\Phi 18 \text{ mm/m}$$

ONE-WAY SOLID SLAB (7.2m x 3.0m) (S1)

GIVEN:-

$$f_y = 420 \text{ MPa} \quad f_c = 35 \text{ MPa} \quad \Phi = 10 \text{ mm}$$

$$W_{dl}(\text{finishing}) = 1.185 \text{ KN/m}^2 \quad WLL = 1.9 \text{ KN/m}^2$$

Both end continuous

$$h_{\min} L/28 = 3000/28 = 107 \text{ mm}$$

$$h_{\text{used}} = 150 \text{ mm}$$

Effective depth :-

$$d = h - c.c. - \Phi/2 = 150 - 25 - 10/2 = 120 \text{ mm}$$

LOADS CALCULATIONS:-

$$W_{dl} = W_{\text{slab}} + W_{\text{finishing}}$$

$$(1 \times 1 \times 0.15) \times 24 + 1.185 = 4.785 \text{ KN/m}^2$$

ULTIMATE LOAD:-

$$U = 1.2(W_{dl}) + 1.6(1.9) = 8.782 \text{ KN/m}^2$$

$$M_u (\text{at } a) = UL^2/9 = 8.782(3^2)/9 = 8.782 \text{ KN.m/m}$$

$$M_n = M_u/0.9 = 8.782/0.9 = 9.757 \text{ KN.m}$$

$$R_n = M_n/1000 \times b \times d^2 = 9.757/1000 \times 1 \times (0.12^2) = 0.677 \text{ MPa}$$

$$m = f_y/0.85f_c = 420/0.85 \times 35 = 14.118$$

$$\rho = \frac{1}{m} \left[1 - \sqrt{1 - \frac{2mR_n}{f_y}} \right]$$

$$\rho = 1/14.118(1 - \sqrt{1 - 2 \times 14.118 \times 0.677/420}) = 1.63 \times 10^{-3}$$

STEEL AREA

$$A_s = (1.63 \times 10^{-3}) \times 1000 \times 120 = 195 \text{ mm}^2$$

$$A_b = \pi d^2 / 4$$

$$= \pi (8)^2 / 4 = 50 \text{ mm}^2$$

$$\text{Or } = \pi (6)^2 / 4 = 28.3 \text{ mm}^2$$

$$\#n.\text{bars} = A_s / A_b = 195 / 28.3 = 6.89 = 7\Phi 6 \text{ mm/m}$$

$$M_u (\text{at } b) = u l^2 / 14 = 8.782 (3^2) / 14 = 5.64 \text{ KN.m/m}$$

$$M_n = M_u / 0.9 = 5.64 / 0.9 = 6.27 \text{ KN.m}$$

$$R_n = 6.27 / 1000 \times 1 \times (0.12^2) = 0.435 \text{ MPa}$$

$$\rho_{+ve} = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times 0.435 / 420}) = 1.04 \times 10^{-3}$$

$$\rho_{min} = 1.4 / f_y = 1.4 / 420 = 3.3 \times 10^{-3}$$

$$\rho_{+ve} < \rho_{min}$$

STEEL AREA

$$A_s = \rho_{min} \cdot b \cdot d = 3.3 \times 10^{-3} \times 1000 \times 120 = 400 \text{ mm}^2$$

Use $\Phi = 10$

$$A_b = \pi d^2 / 4$$

$$\pi (10)^2 / 4 = 78.5 \text{ mm}^2$$

$$\#n.\text{bars} = A_s / A_b = 400 / 78.5 = 5\Phi 10 \text{ mm/m}$$

TWO-WAY RIBBED SLAB (9.0m x 7.2m) (R2)

GIVEN:-

$$f_y = 420 \text{ MPa} \quad f_c = 35 \text{ MPa} \quad \Phi = 18 \text{ mm}$$

$$\Phi_{\text{stirrups}} = 10 \text{ mm} \quad WLL = 1.9 \text{ KN/m}^2$$

To find h

$$h = \frac{\ln(0.8 + \frac{f_y}{1500})}{36 + 5\beta(\alpha m - 0.2)}$$

Assumed $\alpha m = 2$

$$\beta = 9.0/7.2 = 1.25$$

$$h = 9 (0.8 + 420/1500) / 36 + 5 \times 1.25 (2 - 0.2) = 205 \text{ mm}$$

$$h_{\text{used}} = 300 \text{ mm}$$

effective depth

$$d = h - c - \Phi/2 - \Phi_{\text{stirrups}} = 300 - 10 - 9 - 10 = 271 \text{ mm}$$

LOAD CALCULATIONS:-

Wdl =

$$\text{Upper slab (solid)} = (1 \times 1 \times 0.1) \times 24 = 2.4 \text{ KN/m}^2$$

$$\text{Ribs (x-direction)} = 2 \times (0.1 \times 0.2 \times 1) \times 24 = 0.96 \text{ KN/m}^2$$

$$\text{Ribs (y-direction)} = 2 \times (0.1 \times 0.2 \times 1) \times 24 = 0.96 \text{ KN/m}^2$$

$$\text{Blocks} = (8 \times 2) \times 0.15 = 2.4 \text{ KN/m}^2$$

$$\text{Slab self-weight per } 1 \text{ m}^2 = 6.72 \text{ KN/m}^2$$

Finishing =

$$\text{Porcelain} = (1 \times 1 \times 0.005) \times 21 = 0.105 \text{ KN/m}^2$$

$$\text{Mortar} = (1 \times 1 \times 0.05) \times 20 = 1 \text{ KN/m}^2$$

$$\text{Gypsum board ceiling} = (1 \times 1 \times 0.02) \times 4.4 = 0.08 \text{ KN/m}^2$$

$$\text{Total finishing load per } 1\text{m}^2 = 1.185 \text{ KN/m}^2$$

$$\text{TOTAL DEAD LOAD} = 6.72 + 1.185 = 7.905 \text{ KN/m}^2$$

ULTIMATE LOAD:

$$1.2(\text{WDL}) + 1.6(\text{WLL}) = 1.2(7.905) + 1.6(1.9) = 12.526 \text{ KN/m}^2$$

-UL PER RIB:

$$\text{UL/RIB} = 12.526/2 = 6.263 \text{ KN/m}^2$$

CALCULATION OF MOMENT:-

$$-M_u = CUL^2 = 0.033 \times 6.263(9^2) = 16.74 \text{ KN/RIB}$$

$$-M_n = M_u / 0.9 = 16.74 / 0.9 = 18.6 \text{ KN/RIB}$$

STEEL RATIO:-

$$-R_n = M_n / 1000 \times b(d^2) = 18.6 / 1000 \times 1 \times (0.271^2) = 0.253 \text{ MPa}$$

$$-m = f_y / 0.85f_c = 420 / 0.85 \times 35 = 14.118$$

$$\rho = \frac{1}{m} \left[1 - \sqrt{1 - \frac{2mR_n}{f_y}} \right]$$

$$\rho = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times 0.253 / 420}) = 6.05 \times 10^{-4}$$

$$\rho_{\min} = 1.4 / f_y = 1.4 / 420 = 3.33 \times 10^{-3}$$

STEEL AREA

$$A_s = \rho b d = 3.33 \times 10^{-3} \times 500 \times 271 = 451.2$$

$$A_b = \pi d^2 / 4$$

$$= \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

#n.bars

$$A_s / A_b$$

$$451.2 / 254.5 = 1.77 = 2\Phi 18 \text{ mm/m}$$

Design Drop Beam (DB1):-

Deflection requirements:

$$h_{\min} = \text{span}/21$$

$$h_{\min} = 7.2/21 = 343\text{mm}$$

$$h_{\text{used}} = 600\text{mm}, W = 300\text{mm}$$

Load calculation:-

Beam self-weight=

$$0.6 \times 0.3 \times 1 \times 24 = 4.32 \text{ KN/m}$$

Wall load=

$$(0.25 \times 1 \times 1) \times 1.5 \times 12.5 = 4.7 \text{ KN/m}$$

Slab load =

$$7.905 \text{ KN/m}^2$$

Total dead load=

$$4.32 + 4.7 + 7.905 = 16.925 \text{ KN/m}$$

Design load:-

$$U = 1.2(W_{dl}) + 1.6(W_{ll}) = \text{KN/m}^2$$

$$= 1.2(16.925) + 1.6(1.9) = 23.35 \text{ KN/m}^2$$

$$M_u(a) = UL^2/24 = 23.35 \times (7.2)^2/24 = 50.4 \text{ KN.m}$$

$$M_u(b) = UL^2/14 = 23.35 \times (7.2)^2/14 = 86.5 \text{ KN.m}$$

$$M_u(c) = UL^2/10 = 23.35 \times (7.2)^2/10 = 121 \text{ KN.m}$$

$$M_u(d) = UL^2/16 = 23.35 \times (7.2)^2/16 = 75.65 \text{ KN.m}$$

$$M_u(e) = UL^2/11 = 23.35 \times (7.2)^2/11 = 110 \text{ KN.m}$$

$$M_n = M_u / 0.9$$

$$M_n (a) = 50.4 / 0.9 = 56 \text{ KN.m}$$

$$M_n (b) = 86.5 / 0.9 = 96 \text{ KN.m}$$

$$M_n (c) = 121 / 0.9 = 134.4 \text{ KN.m}$$

$$M_n (d) = 75.65 / 0.9 = 84 \text{ KN.m}$$

$$M_n (e) = 110 / 0.9 = 122.2 \text{ KN.m}$$

Effective depth:-

$$d = h - c - \Phi / 2 - \Phi_{\text{stirrups}} = 600 - 20 - 18 / 2 - 10 = 561 \text{ mm}$$

Steel Ratio:-

$$m = f_y / 0.85 f_c = 420 / 0.85 \times 35 = 14.118$$

$$R_n = M_n \times 10^6 / b d^2$$

$$R_n (a) = 56 \times 10^6 / 300 \times (561)^2 = 0.59 \text{ MPa}$$

$$R_n (b) = 96 \times 10^6 / 300 \times (561)^2 = 1.0 \text{ MPa}$$

$$R_n (c) = 134.4 \times 10^6 / 300 \times (561)^2 = 1.4 \text{ MPa}$$

$$R_n (d) = 84 \times 10^6 / 300 \times (561)^2 = 0.89 \text{ MPa}$$

$$R_n (e) = 122.2 \times 10^6 / 300 \times (561)^2 = 1.3 \text{ MPa}$$

$$\rho = \frac{1}{m} \left[1 - \sqrt{1 - \frac{2mR_n}{f_y}} \right]$$

$$\rho (a) = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times (0.59) / 420}) = 1.42 \times 10^{-3}$$

$$\rho (b) = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times (1.0) / 420}) = 2.42 \times 10^{-3}$$

$$\rho (c) = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times (1.4) / 420}) = 3.4 \times 10^{-3}$$

$$\rho (d) = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times (0.89) / 420}) = 2.15 \times 10^{-3}$$

$$\rho (e) = 1 / 14.118 (1 - \sqrt{1 - 2 \times 14.118 \times (1.3) / 420}) = 3.1 \times 10^{-3}$$

$$\rho_{\min} = 1.4 / f_y = 1.4 / 420 = 3.33 \times 10^{-3}$$

ρ all points but (c) $< \rho_{\min}$

Steel Ratio:-

$$A_s = \rho b d$$

$$A_s(c) = (3.4 \times 10^{-3})(300)(561) = 572.2 \text{ mm}^2$$

$$A_s(\text{other points}) = (3.33 \times 10^{-3})(300)(561) = 555.4 \text{ mm}^2$$

$$A_b = \pi d^2 / 4$$

$$= \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

$$\#n.\text{bars (c)} = A_s / A_b = 572.2 / 254.5 = 2.25 = 3 \text{ bars}$$

$$\#n.\text{bars}(\text{other points}) = A_s / A_b = 555.4 / 254.5 = 2.2 = 3 \text{ bars}$$

Spacing between bars:-

$$S = b - 2c - 2\Phi_{\text{bar}} - 2\Phi_{\text{stirrups}} / (n-1) =$$

$$300 - 2 \times 20 - 2 \times 18 - 2 \times 10 / 3 - 1 = 102 \text{ mm}$$

Design Hidden Beam(HB1):-

Deflection requirements:

$$h_{\min} = \text{span}/21$$

$$h_{\min} = 7.2/21 = 342.85\text{mm}$$

$$h_{\text{used}} = 300\text{mm}, W = 600\text{mm}$$

Load calculation:-

Beam self-weight=

$$0.3 \times 0.6 \times 1 \times 24 = 4.32 \text{ KN/m}$$

Wall load=

$$(0.2 \times 1 \times 1) \times 1.5 \times 12.5 = 3.75 \text{ KN/m}$$

Slab load =

$$7.905 \text{ KN/m}^2$$

Total dead load=

$$4.32 + 3.75 + 7.905 = 15.975 \text{ KN/m}$$

Design load:-

$$U = 1.2(W_{dl}) + 1.6(W_{ll}) = \text{KN/m}^2$$

$$= 1.2(15.975) + 1.6(1.9) = 22.2 \text{ KN/m}^2$$

$$M_u(a) = UL^2/16 = 22.2 \times (7.2)^2/16 = 71.3 \text{ KN.m}$$

$$M_u(b) = UL^2/14 = 22.2 \times (7.2)^2/14 = 81.5 \text{ KN.m}$$

$$M_u(c) = UL^2/10 = 22.2 \times (7.2)^2/10 = 114 \text{ KN.m}$$

$$M_u(d) = UL^2/16 = 22.2 \times (7.2)^2/16 = 71.3 \text{ KN.m}$$

$$M_u(e) = UL^2/11 = 22.2 \times (7.2)^2/11 = 103.7 \text{ KN.m}$$

$$M_n = M_u / 0.9$$

$$M_n (a) = 71.3 / 0.9 = 79.2 \text{ KN.m}$$

$$M_n (b) = 81.5 / 0.9 = 90.6 \text{ KN.m}$$

$$M_n (c) = 114 / 0.9 = 126.7 \text{ KN.m}$$

$$M_n (d) = 71.3 / 0.9 = 79.2 \text{ KN.m}$$

$$M_n (e) = 103.7 / 0.9 = 115.2 \text{ KN.m}$$

Effective depth:-

$$d = h - c - \Phi / 2 - \Phi_{\text{stirrups}} = 300 - 20 - 18 / 2 - 10 = 261 \text{ mm}$$

Steel Ratio:-

$$m = f_y / 0.85 f_c = 420 / 0.85 \times 35 = 14.118$$

$$R_n = M_n \times 10^6 / b d^2$$

$$R_n (a) = 79.2 \times 10^6 / 300 \times (261)^2 = 3.9 \text{ MPa}$$

$$R_n (b) = 90.6 \times 10^6 / 300 \times (261)^2 = 4.43 \text{ MPa}$$

$$R_n (c) = 126.7 \times 10^6 / 300 \times (261)^2 = 6.2 \text{ MPa}$$

$$R_n (d) = 79.2 \times 10^6 / 300 \times (261)^2 = 3.9 \text{ MPa}$$

$$R_n (e) = 115.2 \times 10^6 / 300 \times (261)^2 = 5.6 \text{ MPa}$$

$$\rho = \frac{1}{m} \left[1 - \sqrt{1 - \frac{2mR_n}{f_y}} \right]$$

$$\rho (a) = 1 / 14.118 \left(1 - \sqrt{1 - 2 \times 14.118 \times (3.9) / 420} \right) = 1 \times 10^{-2}$$

$$\rho (b) = 1 / 14.118 \left(1 - \sqrt{1 - 2 \times 14.118 \times (4.43) / 420} \right) = 0.011$$

$$\rho (c) = 1 / 14.118 \left(1 - \sqrt{1 - 2 \times 14.118 \times (6.2) / 420} \right) = 0.016$$

$$\rho (d) = 1 / 14.118 \left(1 - \sqrt{1 - 2 \times 14.118 \times (3.9) / 420} \right) = 1 \times 10^{-2}$$

$$\rho (e) = 1 / 14.118 \left(1 - \sqrt{1 - 2 \times 14.118 \times (5.6) / 420} \right) = 0.015$$

$$\rho_{\min} = 1.4 / f_y = 1.4 / 420 = 3.33 \times 10^{-3}$$

ρ all points $> \rho_{\min}$

Steel Ratio:-

$$A_s = \rho b d$$

$$A_s (a) = (1 \times 10^{-2})(300)(261) = 783 \text{ mm}^2$$

$$A_s (b) = (0.011)(300)(261) = 861.3 \text{ mm}^2$$

$$A_s (c) = (0.016)(300)(261) = 1252.8 \text{ mm}^2$$

$$A_s (d) = (1 \times 10^{-2})(300)(261) = 783 \text{ mm}^2$$

$$A_s (e) = (0.015)(300)(261) = 1174.5 \text{ mm}^2$$

$$A_b = \pi(18)^2 / 4 = 254.5 \text{ mm}^2$$

$$\#n.\text{bars}(a) = A_s / A_b = 783 / 254.5 = 3.1 = 4 \text{ bars}$$

$$\#n.\text{bars}(b) = A_s / A_b = 861.3 / 254.5 = 3.38 = 4 \text{ bars}$$

$$\#n.\text{bars}(c) = A_s / A_b = 1252.8 / 254.5 = 4.92 = 5 \text{ bars}$$

$$\#n.\text{bars}(d) = A_s / A_b = 783 / 254.5 = 3.07 = 4 \text{ bars}$$

$$\#n.\text{bars}(e) = A_s / A_b = 1174.5 / 254.5 = 4.61 = 5 \text{ bars}$$

Spacing between bars:-

$$S = b - 2c - 2\Phi_{\text{bar}} - 2\Phi_{\text{stirrups}} / (n-1)$$

$$S(a) = 600 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (4-1) = 168 \text{ mm}$$

$$S(b) = 600 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (4-1) = 168 \text{ mm}$$

$$S(c) = 600 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (5-1) = 126 \text{ mm}$$

$$S(d) = 600 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (4-1) = 168 \text{ mm}$$

$$S(e) = 600 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (5-1) = 126 \text{ mm}$$

DESIGN OF C1:-

LOAD CALCULATION:

$$DB1 = 4.32 \times 3.6 = 15.5 \text{ KN}$$

$$DB1 = 4.32 \times 3.6 = 15.5 \text{ KN}$$

$$DB2 = 4.32 \times 3 = 13 \text{ KN}$$

$$\text{-TOTAL BEAMS SELF-WEIGHT= 44 KN}$$

$$S1 = 7.905 \times (3.6 \times 3) = 85.4 \text{ KN}$$

$$S2 = 7.905 \times (4.5 \times 3.6) = 128 \text{ KN}$$

$$S3 = 4.785 \times (3.6 \times 1.5) \times 2 = 51.7 \text{ KN}$$

$$\text{-TOTAL SLAB SELF-WEIGHT= 265 KN}$$

$$\text{Two walls on (DB1)= } 4.7 \times (3.6 \times 3.3) \times 2 = 111.7 \text{ KN}$$

$$\text{Wall on (DB2)= } 3.75 \times (3 \times 3.3) = 37.1 \text{ KN}$$

$$\text{-TOTAL WALL WEIGHT= 148.8 KN}$$

$$\text{PLASTER ON WALL 1\&1 = } 0.4 \times (3.6 \times 3.6) \times 2 \times 2 = 19 \text{ KN}$$

$$\text{PLASTER ON WALL 2 = } 0.4 \times (3 \times 3.3) \times 2 = 8 \text{ KN}$$

$$\text{-TOTAL PLASTER WEIGHT= 27 KN}$$

$$\text{-ADDITIONAL WEIGHT= 2 KN}$$

$$\text{-COLUMN SELF-WEIGHT= } 0.5 \times 0.3 \times 24 \times 3.9 = 14.04$$

TOTAL LOAD ON

$$(C1) = 44 + 265 + 148.8 + 27 + 2 + 14.04 = 500.84 \text{ KN}$$

DESIGN LOAD:-

$$-U = 1.2(500.84) + 1.6(1.9) = 604.05 \text{ KN}$$

$$-\text{TOTAL LOAD FOR TWO FLOORS} = 604.05 \times 2 = 1208.1 \text{ KN}$$

$$f_c = 35 \text{ MPa} \quad f_y = 420 \text{ MPa} \quad \rho_g = 2\% = 0.02$$

$$-P_u = 0.8 \times 0.7 (0.85 f_c (1 - \rho_g) + \rho_g f_y) A_g$$

$$P_u = 0.8 \times 0.7 (0.85 (35) (1 - 0.02) + (0.02 \times 420)) A_g$$

$$1208.1 = 21 A_g$$

$$A_g = 1208.1 / 21 = 0.057 \text{ m}^2 = 570 \text{ cm}^2 = 25 \times 22.8$$

$$A_{g \text{ used}} = 50 \times 30 = 1500 \text{ cm}^2$$

$$A_s = \rho_g \times A_g = 0.02 \times 1500 = 30 \text{ cm}^2 = 3000 \text{ mm}^2$$

$$A_b = \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

$$\#n.\text{bars} = A_s / A_b = 3000 / 254.5 = 11.78 = 12 \text{ bars}$$

$$S = b - 2c - 2\Phi_{\text{bar}} - 2\Phi_{\text{stirrups}} / (n - 1)$$

$$S = 500 - 2 \times 20 - 2 \times 18 - 2 \times 10 / (12 - 1) = 36.7 \text{ mm} = 0.037 \text{ m}$$

DESIGN OF C2:-

LOAD CALCULATION:

$$DB1 = 4.32 \times 3.6 = 15.5 \text{ KN}$$

$$DB1 = 4.32 \times 3.6 = 15.5 \text{ KN}$$

$$DB2 = 4.32 \times 3 = 13 \text{ KN}$$

$$\text{-TOTAL BEAMS SELF-WEIGHT} = 44 \text{ KN}$$

$$S1 = 7.905 \times (3.6 \times 3) \times 2 = 170.7 \text{ KN}$$

$$\text{-TOTAL SLAB SELF-WEIGHT} = 170.7 \text{ KN}$$

$$\text{Two walls on (DB1)} = 4.7 \times (3.6 \times 3.3) \times 2 = 111.7 \text{ KN}$$

$$\text{Wall on (DB2)} = 3.75(3 \times 3.3) = 37.1 \text{ KN}$$

$$\text{-TOTAL WALL WEIGHT} = 148.8 \text{ KN}$$

$$\text{CLADDING ON WALL 1\&1} = 0.1(3.6 \times 3.9) \times 2 = 2.8 \text{ KN}$$

$$\text{PLASTER ON WALL 1\&1} = 0.4(3.6 \times 3.6) \times 2 = 10.4 \text{ KN}$$

$$\text{PLASTER ON WALL 2} = 0.4(3 \times 3.6) \times 2 = 8.6 \text{ KN}$$

$$\text{-TOTAL CLADDING \& PLASTER WEIGHT} = 21.2 \text{ KN}$$

$$\text{-ADDITIONAL WEIGHT} = 2 \text{ KN}$$

$$\text{-COLUMN SELF-WEIGHT} = 0.4 \times 0.3 \times 24 \times 3.9 = 11.2$$

TOTAL LOAD ON

$$(C2) = 44 + 170.7 + 148.8 + 21.2 + 2 + 11.2 = 398 \text{ KN}$$

DESIGN LOAD:-

$$-U = 1.2(398) + 1.6(1.9) = 480.64 \text{ KN}$$

$$-\text{TOTAL LOAD FOR TWO FLOORS} = 480.64 \times 2 = 961.3 \text{ KN}$$

$$f_c = 35 \text{ MPa} \quad f_y = 420 \text{ MPa} \quad \rho_g = 2\% = 0.02$$

$$-P_u = 0.8 \times 0.7 (0.85 f_c (1 - \rho_g) + \rho_g f_y) A_g$$

$$P_u = 0.8 \times 0.7 (0.85 (35) (1 - 0.02) + (0.02 \times 420)) A_g$$

$$961.3 = 21 A_g$$

$$A_g = 961.3 / 21 = 0.045 \text{ m}^2 = 450 \text{ cm}^2 = 20 \times 22.5$$

$$A_{g \text{ used}} = 40 \times 30 = 1200 \text{ cm}^2$$

$$A_s = \rho_g \times A_g = 0.02 \times 1200 = 24 \text{ cm}^2 = 2400 \text{ mm}^2$$

$$A_b = \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

$$\#n.\text{bars} = A_s / A_b = 2400 / 254.5 = 9.43 = 10 \text{ bars}$$

$$S = b - 2cc - 2\Phi bar - 2\Phi stirrups / n - 1$$

$$S = 400 - 2 \times 20 - 2 \times 18 - 2 \times 10 / 10 - 1 = 34 \text{ mm} = 0.034 \text{ m}$$

DESIGN OF C3:-

LOAD CALCULATION:

$$DB1 = 4.32 \times 3.6 = 15.5 \text{ KN}$$

$$DB3 = 4.32 \times 1.5 = 6.5 \text{ KN}$$

$$\text{-TOTAL BEAMS SELF-WEIGHT} = 22 \text{ KN}$$

$$S1 = 7.905 \times (3.6 \times 3) = 85.4 \text{ KN}$$

$$S2 = 7.905 \times (4.5 \times 3.6) = 128 \text{ KN}$$

$$\text{-TOTAL SLAB SELF-WEIGHT} = 213.4 \text{ KN}$$

$$\text{Two walls on (DB1)} = 4.7 \times (3.6 \times 3.3) = 55.8 \text{ KN}$$

$$\text{Wall on (DB2)} = 4.7(1.5 \times 3.3) = 23.3 \text{ KN}$$

$$\text{-TOTAL WALL WEIGHT} = 79.1 \text{ KN}$$

$$\text{CLADDING ON WALL 1} = 0.1(3.6 \times 3.9) = 1.4 \text{ KN}$$

$$\text{PLASTER ON WALL 2} = 0.4(3.6 \times 3.6) = 5.2 \text{ KN}$$

$$\text{CLADDING ON WALL 2} = 0.1(1.5 \times 3.9) = 0.6 \text{ KN}$$

$$\text{PLASTER ON WALL 2} = 0.4(3 \times 3.6) \times 2 = 2.2 \text{ KN}$$

$$\text{-TOTAL CLADDING WEIGHT} = 9.4 \text{ KN}$$

$$\text{-ADDITIONAL WEIGHT} = 2 \text{ KN}$$

$$\text{-COLUMN SELF-WEIGHT} = 0.3 \times 0.3 \times 24 \times 3.9 = 8.4$$

$$\text{TOTAL LOAD ON (C3)} = 22 + 213.4 + 79.1 + 9.4 + 2 + 8.4 = 334.3 \text{ KN}$$

DESIGN LOAD:-

$$-U = 1.2(334.3) + 1.6(1.9) = 404.2 \text{ KN}$$

$$-\text{TOTAL LOAD FOR TWO FLOORS} = 404.2 \times 2 = 808.4 \text{ KN}$$

$$f_c = 35 \text{ MPa} \quad f_y = 420 \text{ MPa} \quad \rho_g = 2\% = 0.02$$

$$-P_u = 0.8 \times 0.7 (0.85 f_c (1 - \rho_g) + \rho_g f_y) A_g$$

$$P_u = 0.8 \times 0.7 (0.85 (35) (1 - 0.02) + (0.02 \times 420)) A_g$$

$$808.4 = 21 A_g$$

$$A_g = 808.4 / 21 = 0.038 \text{ m}^2 = 380 \text{ cm}^2 = 20 \times 19$$

$$A_{g \text{ used}} = 30 \times 25 = 750 \text{ cm}^2$$

$$A_s = \rho_g \times A_g = 0.02 \times 750 = 15 \text{ cm}^2 = 1500 \text{ mm}^2$$

$$A_b = \pi (18)^2 / 4 = 254.5 \text{ mm}^2$$

$$\#n.\text{bars} = A_s / A_b = 1500 / 254.5 = 5.89 = 6 \text{ bars}$$

$$S = b - 2cc - 2\Phi bar - 2\Phi stirrups / n - 1$$

$$S = 300 - 2 \times 20 - 2 \times 18 - 2 \times 10 / 6 - 1 = 40.8 \text{ mm} = 0.04 \text{ m}$$